Solving a generalisation of tic-tac-toe or k-in-a-row game on infinite $\{m, n\}$ tilings of 2D space.

Subtitle

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1 Introduction

1.1 Insights

In $\{m, n\}$ tiling let

$$m = 2i : i \in \mathbb{N}. \tag{1}$$

It matters so that there ale straight lines going through the shapes. Then in $n = 2j + 1$ tilings, diagonals do not make sense. In $n = 2j$ tilings, however, diagonals make sense. We then distinguish between a $\{m, n = 2j\}_m$ and $\{m, n = 2j\}_{vn}$ configurations named after the Moore (M) and Von Neumann (VN) neighbourhoods. The Moore version permitting diagonals to be used and the Von Neumann version only permitting edge-connected series. The classical tic-tac-toe is then played on a $\{4, 4\}_m$ board. In the spirit of m, n, k games we include also the the length of the winning sequence as follows: $\{m, n\}^k$. The number of axis that you can win on then becomes m in the M configuration and $m/2$ in the VN configuration.

1.2 Notation

1.2.1 Schläfli symbol

The Schläfli symbol is used to compactly denote any regular polytope or tessellation. For our case it will be of the form $\{p, q\}$, where $\{p\}$ is a regular p-sided polygon and q is the number of these polygons joined at each vertex.

1.2.2 Game versions

$(1, 1, 1)$ where each player occupies 1 position in a move. $(1, 2, 2)$ where the first move occupies 1 position and then both players occupy 2 positions per move.

1.3 Maker-Breaker games

In a Maker-Breaker game, the Maker is the player trying to achieve the winning objective and Breaker is the player trying to stop the Maker from achieving this.

1.3.1 Insights into Maker-Maker games

If a k-game is a Breaker win, we know that the original Maker-Maker version is a draw.

1.3.2 Strategy stealing argument for $(1, 1, 1)$

In a Maker-Maker game, if there was a strategy such that P2 would have a winning strategy, P1 could just steal the strategy in a way where P1 would make a random move and then effectively play the role of P2 and employ that winning strategy. This is a contradiction as both P1 and P2 have a winning strategy. Therefore only P1 can have a winning strategy.

2 Tiling spaces

No spherical tilings since they are all finite and do not form an infinite playing board.

The proof for any tiling with $k = 1$ is trivial as the first player wins by making any move.

2.1 Euclidean tilings

The only regular tilings of Euclidean space are $\{2, \infty\}, \{3, 6\}, \{4, 4\}, \{6, 3\}, \{\infty, 2\}$. We do not look at the $\{\infty, 2\}$ tiling as it does not provide an infinite playing board. We also exclude the $\{3, 6\}$ tiling as it does not satisfy condition 1. Finally, we also do not spend too much time looking at the $\{4, 4\}$ tiling as this is the case with regular m, n, k-games and has been studied extensively. We are left with the $\{2, \infty\}$ and $\{6, 3\}$ tilings, which we take a close look at.

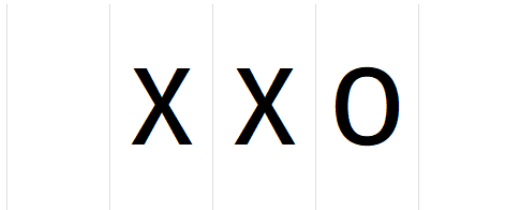

Figure 1: Drawing of the winning strategy for $\{2, \infty\}^2$.

2.1.1 The $\{2, \infty\}$ tiling

This is a very trivial case as it is a win for P1 in the case of $k \leq 2$ and a draw in the case $k > 2$.

For the case $k = 2$ P1 has a winning strategy. We can see that the $\{2, \infty\}$ has a reflective symmetry. Therefore the is only one game (up to symmetry) of $\{2, \infty\}^2$. It is shown in Figure 1.

For the case $k > 2$ We can show this using a simple pairing argument where we tile the space as in Figure 2.

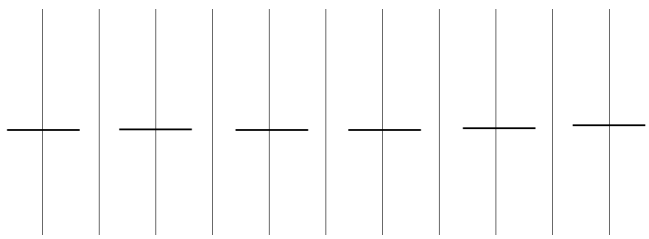

Figure 2: Drawing of the pairing strategy for $\{2, \infty\}^3$.

NEEDS PROOF OF SAME THING FOR $(1, 2, 2)$

2.1.2 The $\{6, 3\}$ tiling

Since the vertex degree in this tiling is odd, we will not consider diagonals.

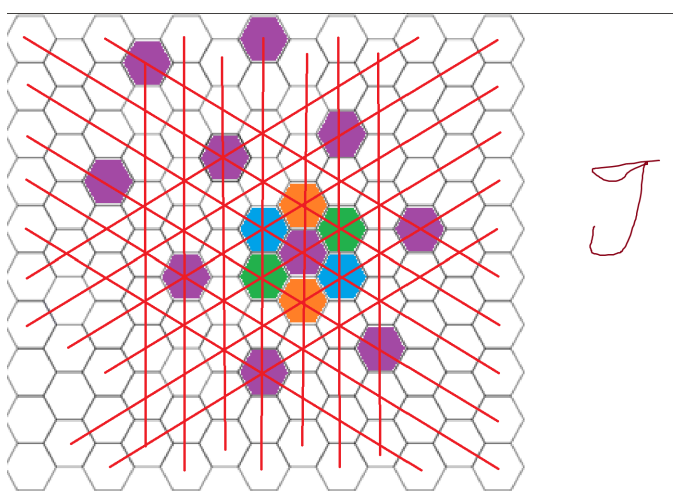

Figure 3: Drawing of the pairing strategy for $\{6, 3\}^9$.

Figure 3 shows a winning breaker strategy for the $\{6, 3\}^9$ game.
NEEDS PROOF FOR $k < 5$**.**

NEEDS PROOFS FOR $(1, 2, 2)$

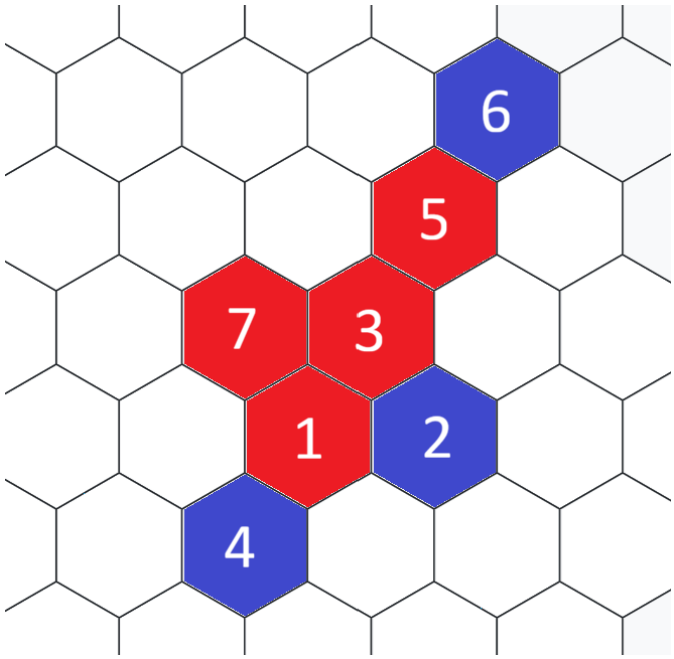

Figure 4: Forced Winning configuration (up to symmetry) for $\{6,3\}^4$.

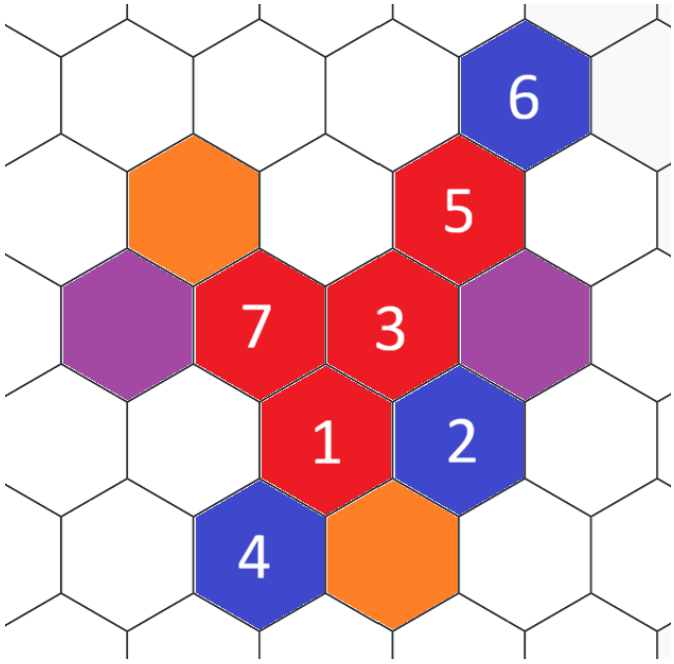

Figure 5: Threat illustration for $\{6,3\}^4$.

2.2 Hyperbolic tilings

There is an infinite amount of hyperbolic tilings. We will therefore look at some representative examples and then draw general conclusions.

References